

SV-16 microcontroller — PCB design blueprint

Status: design specification for review — no board has been built or ordered. Nothing in this document has been validated on hardware; every number is either taken from a cited document (Lattice datasheet / technical note / BSDL model) or measured from this repository's own build artifacts. Items that still need bench or vendor verification are collected in §13.2.

What it is: a complete component list, electrical specification and connection diagram for turning the SV-16 soft-core that lives in this repository into a **stand-alone MCU-class board** built around a single Lattice **LFE5U-12F-6TG144C** (ECP5, 144-lead TQFP, speed grade 6, commercial temperature).

Sourcing note (revision 1.1): the component list was originally written for US/EU distributors. It has been re-sourced for India: every part below is available from a domestic vendor, with the electrical equivalent spelled out where the original choice had to be replaced. §3.6 is the change list.

It is not: a copy of an evaluation kit, and not a general-purpose FPGA dev board. The board exists to run the firmware in `firmware/` — boot from non-volatile memory, talk on a serial console, be reprogrammed in the field, and drive a small motor — which is exactly what the SoC in `rtl/` implements.

Device	LFE5U-12F-6TG144C — ECP5, 12,144 LUT4, 32 × sysMEM (72 KB), 28 multipliers, 2 PLL, 144-TQFP
Fit of the current SoC	9,191 LUT4 (75.7 %), 4,772 FF (39.3 %), 18/32 DP16KD (56.2 %), 1/2 EHXPLLL, 52/98 I/O
Bitstream	292,752 B compressed (<code>build/pll/sv16_top.bit</code>), 45.45 MHz max core clock
I/O banks used	0, 1, 2, 3, 7, 8 — all LVCMOS33 (3.3 V), 52 signals constrained, 46 spare I/O
Storage	2 × 3.3 V SPI NOR: one for the FPGA bitstream, one for SV-16 firmware images
Console / update	USB-C → CH340C → 115200 8N1 UART → ROM monitor (erase / upload / verify / boot)
Debug / config	1×10 0.1" and 2×5 0.05" JTAG headers; TCK ≤ 25 MHz (BSDL <code>TAP_SCAN_CLOCK</code>)
Power in	7–12 V DC barrel (recommended) or USB-C 5 V; 3.3 V / 2.5 V / 1.1 V rails generated on board
Sourcing	every part is orderable from an Indian vendor (Robu, Zbotic, QuartzComponents, Sunrom, Hubtronics, Sharvie, KTRON, iFutureTech, DNA Tech) except the FPGA and the oscillator, which come from digikey.in / mouser.in / element14 India — see §3

0.1 Where the numbers come from

Fact	Source
Pin numbers, banks, pad names, the 98 user-I/O pins	<code>board/ecp5_tqfp144.json</code> , cross-built from the prjtrellis device database for LFE5U-12F TQFP144 (98 user-I/O pin numbers + banks) and the Lattice BSDL model FPGA-MD-02097 (LFE5U_25F_XXTG144, pad names, the 7 dedicated sysCONFIG pins, the 4 TAP pins, every VCC/VCCAUX/VCCIO/VSS pin)
Which pin carries which design signal	<code>constraints/ecp5_144tqfp.lpf</code> — the same file the bitstream is built with

Fact	Source
Rail voltages, tolerances, ramp rates, POR trip points, LVCMOS33 range	ECP5 and ECP5-5G.pdf in this repository — FPGA-DS-02012-3.4 , Tables 3.1–3.5, 3.8, 3.11, §4.3.2
Configuration strapping, pull-ups, MCLK, PERSISTENT	Lattice TN1260 (ECP5 sysCONFIG) and TN-02039 (sysCONFIG usage guide); pull-up values from the ECP5 hardware checklist TN-02038-2.0 Table 6.2
JTAG header pinout	Lattice ECP5 Versa EB98/EB103 Appendix A (1×10 header) and the Digilent/FT2232H 2×5 0.05" standard
Pinout of every constrained signal on the board	board/TQFP144_PINOUT.md (generated)

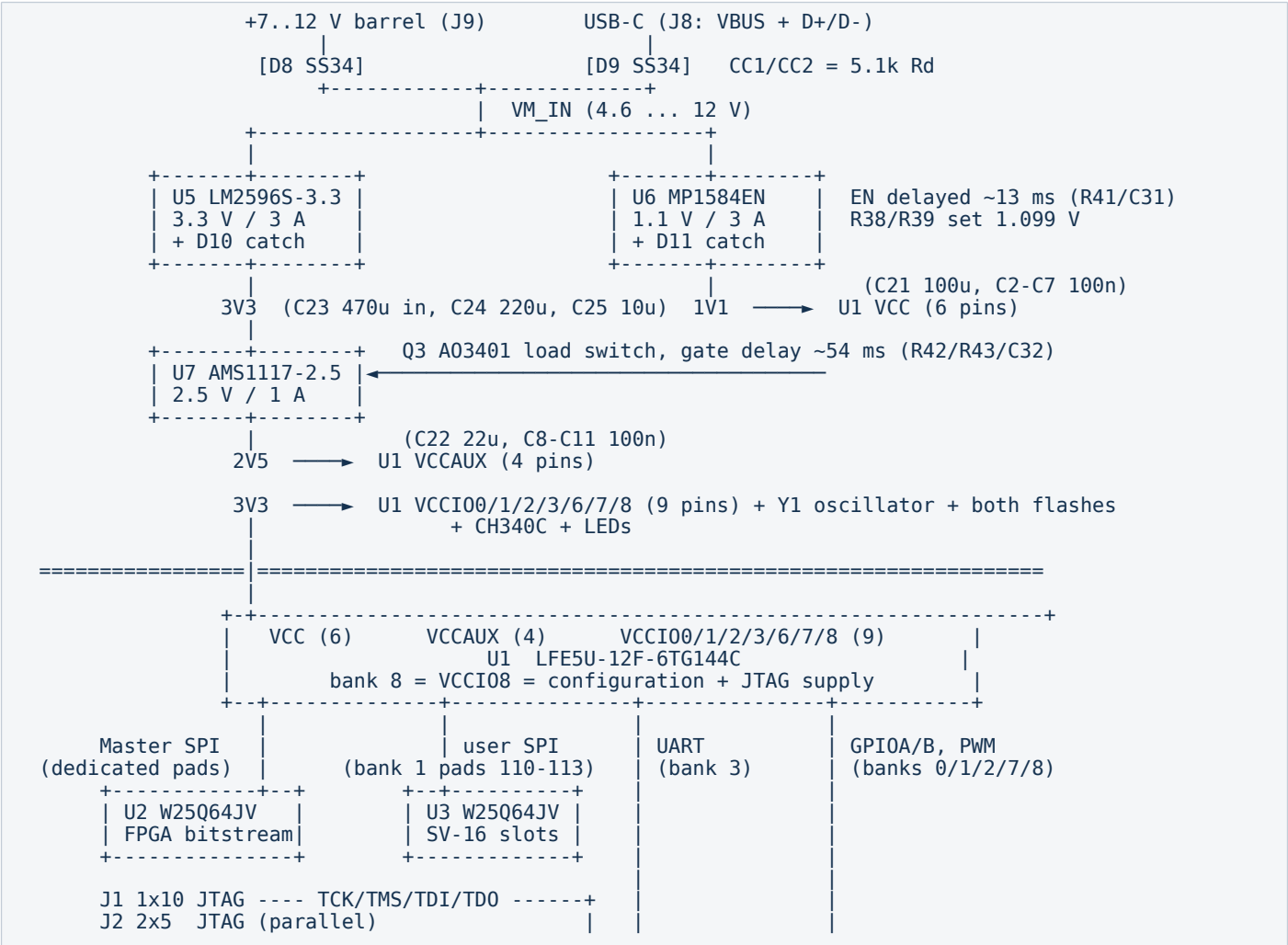
Printables. This document exists twice: BOARD.md (the source of truth, with the generated netlist appendix in board/TQFP144_PINOUT.md) and BOARD.pdf — the same content typeset for printing, with the ASCII diagrams kept in a monospaced face. Rebuild the PDF with `make board-pdf` (python3 -m pip install markdown xhtml2pdf once; nothing else in the build needs those packages).

Drift control. board/TQFP144_PINOUT.md is generated from the two files above and is verified in `make lint`:

```
scripts/sv16_board_pins.py      # regenerate the pin/net table
scripts/sv16_board_pins.py --check  # fail if the checked-in table is stale
scripts/sv16_board_pins.py --summary # bank/usage summary only
```

If somebody moves a signal in constraints/
ecp5_144tqfp.lpf, --check fails until the board document is regenerated, so the schematic and the bitstream cannot silently disagree.

1. Board-level block diagram



```

SW1 reset -> ext_rst_n (pin 134)
SW2 PROGRAMN (pin 57)
D1 DONE, D6 INITN, D2-D5 user LED (GPIOA[3:0])

                                U4 CH340C --+---+ J10 UART hdr
                                (USB console)

J5 EXP-C (GPIOA[7:0]) J3 EXP-A (SPI0 + GPIOA[15:8]) J4 EXP-B (GPIOB[15:0])
J6 motor (PWM, DIR1, DIR2, FAULTn, VM) J7 spare-I/
0 breakout (46 pins, not fitted)
=====

```

The soft-core sees this board through its memory map: GPIOA/GPIOB at 0xF010/0xF070, PWM+quadrature motor block at 0xF030, SPI0 at 0xF050, UART0 at 0xF000, watchdog at 0xF080, SPI flash controller at 0xF0A0 (docs/MEMORY_MAP.md, docs/PERIPHERALS.md). Every board signal below maps to one of those registers — the board adds nothing the SoC cannot already drive.

2. Decisions that shaped the board (and why)

- One I/O rail: 3.3 V everywhere.** The existing LPF constrains all 52 signals to LVCMOS33, which needs VCCIO 3.135–3.465 V (datasheet Table 3.11). Rather than split banks across rails (which would force LPF and bitstream changes), all six used banks sit on 3V3. The flash, the USB bridge and the 25 MHz oscillator are all 3.3 V parts, so no level shifting is needed anywhere.
- Power-up order is enforced by two RC-delayed enables, not by hope.** In Master SPI mode the datasheet requires VCCIO8 to be above the flash's input-high threshold before VCC or VCCAUX reach their POR trip points, otherwise PROGRAMN/INITN must be held low until VCCIO8 is valid (§3.5 of FPGA-DS-02012-3.4). The India-available regulators (LM2596, MP1584) do **not** accept a 3.3 V input, so VCC can no longer be derived from VCCIO8 as in revision 1.0. Instead their enable pins are used: the 1.1 V buck gets an RC on EN (≈ 6 ms at 12 V, ≈ 13 ms at 5 V) and the 2.5 V LDO is fed through a P-MOSFET load switch whose gate RC (R42/R43/C32) delays it by ≈ 54 ms from 12 V. 3V3 = VCCIO8 therefore reaches the flash threshold ≈ 0.3 ms after power-up, long before either other rail moves. §4.3 has the numbers, the bench check is bring-up step 3 in §10.
- Two SPI flashes, not one.** The FPGA's configuration port is not memory-mapped: the MSPI pads (CCLK = pin 54, MOSI = pin 47, MISO = pin 46, CSSPIN = pin 49) belong to the configuration logic while PERSISTENT is set, and the soft-core cannot read or write the bitstream flash through them. The SV-16 firmware store must therefore be a separate device on ordinary user I/O (bank 1, pins 110–113), which is what `sv16_flash_ctrl` and the bootloader already expect. Consequences and the pin-sharing rule are spelled out in §6.4 — this is the single most important thing to understand about this board.
- 25 MHz single-ended oscillator, PLL on-chip.** The design is built for `clk_25m` on pin 133 and the PLL recipe CLKI 2 / CLKFB 3 / CLKOP 16 $\rightarrow$ 37.5 MHz core (VCO 600 MHz), giving 45.45 MHz of margin. A 3.2 $\times$ 2.5 mm 4-pin CMOS XO is cheaper and more robust than a crystal on the FPGA's oscillator pad, and it also supports the `CLKSRC=osc` build (25 MHz, no PLL) unchanged.
- USB-UART instead of a second USB device controller.** The console/update path is the same UART the ROM monitor already speaks: a CH340C makes it a single USB-C cable. The optional DTR $\rightarrow$ PROGRAMN transistor lets a host script reconfigure the FPGA, but it is jumper-disableable because DTR polarity differs between host tools.
- JTAG is the primary configuration and recovery path.** A 1 $\times$ 10 0.1" header (Versa-compatible) and a 2 $\times$ 5 0.05" header (FT2232H/HS2-compatible) are wired in parallel, so both a cheap FT2232H adapter and a Digilent-style probe work. JTAG always works even when the configuration flash is blank or corrupt — that is the brick-recovery path.
- Motor interface on the pins the SoC already drives.** `pwm_out` (pin 88, 16 mA), `motor_dir1/2` (GPIOA[4]/[5] mirrored to pins 89/102) and `motor_fault_n` (pin 103, pulled up) go to a keyed 1 $\times$ 6 header with the motor supply passed through, so an external driver board (or a future on-board driver, §14 V3) plugs straight in. No new logic.
- Every spare pin is brought out.** 46 of the 98 I/O pins are unused by the current design; they go to a 2 $\times$ 25 footprint J7 (not fitted by default) with interleaved grounds so the board can grow without a respin.

3. Bill of materials — India-sourced edition

Everything below is orderable from an Indian vendor that ships domestically: no US/EU freight, no customs paperwork on your side. Where revision 1.0 picked a part that only exists on the global distributors' shelves, it has been replaced by an electrical equivalent that is in stock in India — and because two regulators changed, the power tree, the sequencing scheme and the schematic sheet in §4 and §7.1 changed with them. §3.6 is the change list.

Sourcing tiers

Tier	Vendors	Order from here
1 — Indian component shops (₹ pricing, 2-5 day delivery, no customs)	Robu.in · Zbotic.in · QuartzComponents.in · Sunrom.com · Hubtronics.in (the old rareComponents) · Sharvie Electronics · KTRON India · iFutureTech · Robocraze · DNA Tech · ElectronicsComp	passives, connectors, LEDs, MOSFETs, both SPI flashes, CH340C, USBLC6, LM2596S-3.3, MP1584EN, AMS1117-2.5, modules
2 — Indian storefronts of the global distributors (INR pricing, domestic delivery, they handle customs)	digkey.in · mouser.in · in.element14.com · in.rs-online.com	U1 (the FPGA) and the 25 MHz oscillator — the two parts no hobby shop stocks; also anything tier 1 has run out of
3 — marketplaces (AliExpress, IndiaMART, eBay.in, third-party Amazon.in)	use with care	last resort for passives and modules. Never buy the FPGA here — relabelled, refurbished and out-of-spec ECP5 parts are a known problem, and a fake costs you a board and a fortnight

Prices are indicative, single unit, GST included, from a September 2026 stock check — verify before ordering. Passives are **0603** (stock standard in India and hand-solderable; revision 1.0's 0402 was chosen for density and is a bad idea when you are soldering it yourself). Electrolytics are now required (the two bucks are non-synchronous and the LM2596 needs real bulk capacitance).

3.1 Active devices

Ref	Part number	Specification	Package	Qty	Where (indicative ₹)
U1	LFE5U-12F-6TG144C	CECP5 FPGA, 12,144 LUT4, 72 KB sysMEM, 28 mult, 2 PLL, VCC 1.1 V, VCCAUX 2.5 V, speed 6, commercial 0-85 °C Tj	TQFP-144 20×20 mm, 0.5 mm	1	digkey.in ≈ ₹1,650 incl. GST (in stock Sept 2026) · mouser.in · in.element14.com. No substitute exists — §3.4
U2	W25Q64JVSSIQ	64 Mbit (8 MB) SPI NOR, 2.7-3.6 V, standard SPI + dual/quad	SOIC-8 208 mil	1	Sharvie Electronics ₹75 · Hubtronics ₹117 · KTRON · Robu sells a W25Q64 breakout module for ₹77 (§3.5)
U3	W25Q64JVSSIQ	same device — SV-16 firmware slots. W25Q32JVSSIQ (4 MB) or W25Q16JV is already enough: the images are 32 KB per slot	SOIC-8 208 mil	1	as U2

Ref	Part number	Specification	Package	Qty	Where (indicative ₹)
U4	CH340G	USB 2.0 full-speed ↔ UART, 3.3 V, V3 to 3V3 at 3.3 V operation. Needs an external 12 MHz crystal (Y2) + 2 × 22 pF (C36/C37, 33 pF if the crystal is a 20 pF-load type) — the G version has no internal clock; the CH340C is the pin-compatible part that does not need them	SOP-16 150 mil	1	iFutureTech · Hubtronics · Robu. Pairs with Y2 below
U5	LM2596S-3.3 (was AP63203)	buck, 4.5–30 V in, 3.3 V fixed , 3 A, 150 kHz, non-synchronous → needs D10	TO-263-5	1	Robu ₹51 (XBLW) / ₹61 (Silkor) — listed in stock
U6	MP1584EN-LF-Z (was TPS62823)	buck, 4.5–28 V in, adjustable 0.8–20 V, 3 A, up to 1.5 MHz, non-synchronous → needs D11 and R38/R39	SOIC-8E	1	Hubtronics ₹47 (361 in stock) · Sunrom · others
U7	AMS1117-2.5 (was LP5907)	LDO, 2.5 V, 1 A, needs ≥10 µF on the output	SOT-223	1	Robu / QuartzComponents ₹12–20
Y1	YIC OSC25M-3.3I/S3-25T (was ASEMB)	25.000 MHz XO , 3.3 V CMOS, ±25 ppm, enable/disable	3.2×2.5 mm SMD-4	1	digkey.in ₹111 · any 25 MHz 3.3 V active oscillator in 7050/5032 (Amazon.in / eBay.in sell the 5×7 mm 4-pin type) works
Y2	12 MHz quartz crystal (CH340G only)	12 MHz exactly — the chip accepts 11.98–12.02 MHz, so the 25 MHz part cannot be substituted. CL 12–20 pF, 2-pin passive crystal — the CH340G's internal inverter oscillates it. HC49/US through-hole or 3225 SMD both work	HC49/US or SMD-2	1	Robu / Zbotic ₹15–25. Skipped entirely if U4 is a CH340C
D1–D5	LED 0603, green	DONE indicator + four user LEDs (GPIOA[3:0], active low)	0603	5	Robu / Zbotic ₹1–2 each

Ref	Part number	Specification	Package	Qty	Where (indicative ₹)
D6, D7	LED 0603, red	INITN indicator (DNP by default) + 3V3 power LED	0603	2	as above
D8, D9	SS34	Schottky 40 V / 3 A — barrel and USB input ORing	SMA	2	Robu ₹3
D10	SS34	catch diode for the 3V3 LM2596	SMA	1	as above
D11	SS34	catch diode for the 1V1 MP1584	SMA	1	as above
Q1	2N7002	N-MOSFET, PROGRAMN pull-down from DTR (optional auto-reconfigure)	SOT-23	1	Robu / QuartzComponents ₹5
Q2, Q4	BC817 (NPN) / BC807 (PNP) (was MMBT3904/3906)	indicator drivers for DONE and INITN — identical function, the part Indian shops actually stock	SOT-23	2	Zbotic / Robu ₹2 each
Q3	AO3401	P-MOSFET load switch that delays the 2.5 V LDO (sequencing, §4.3)	SOT-23	1	Robu / Zbotic ₹8
U8	USBLC6-2SC6	USB D+/D− ESD protection, 2 channels + VBUS	SOT-23-6	1	Sunrom ₹30 · DNA Tech ₹32
—	SN74LVC1T45 (optional)	level shifter, only if a 5 V UART adapter is used on J10	SOT-23-6	0	Robu / QuartzComponents if needed

3.2 Connectors, switches, protection

Ref	Item	Specification	Qty	Where / notes
J1	1×10 header, 2.54 mm	JTAG / configuration, Versa pinout (§8)	1	Robu ₹10 — the header that definitely exists in India
J2	2×5 box header, 1.27 mm	Digilent / FT2232H-style adapter	1	1.27 mm box headers are patchy in India: if you cannot find one, leave J2 unpopulated and use J1 with a 2.54 mm ribbon; or fit a 2×5 2.54 mm header and use an adapter cable
J3, J4	2×10 headers, 2.54 mm	EXP-A (SPI0 + GPIOA[15:8]) and EXP-B (GPIOB[15:0])	2	Robu ₹15 each
J5	1×10 header, 2.54 mm	EXP-C: GPIOA[7:0]	1	pins 2/3/5/7 share nets with the config flash — §6.4

Ref	Item	Specification	Qty	Where / notes
J6	1×6 keyed header, 2.54 mm	motor: VM, GND, PWM, DIR1, DIR2, FAULTn	1	Robu ₹15
J7	2×25 header, 2.54 mm	spare-I/O breakout (46 signals + 4 GND)	1	footprint only, not fitted
J8	USB-C receptacle 16-pin — or micro-USB B	power + console. With micro-USB, omit R36/R37	1	Robu / Zbotic ₹25 (USB-C) or ₹10 (micro-USB, easier to hand-solder)
J9	DC barrel jack 5.5 / 2.1 mm	centre positive, 7–12 V	1	Robu ₹20
J10	1×4 header, 2.54 mm	3V3, UART_TX, UART_RX, GND (console without the CH340C)	1	Robu ₹5
SW1, SW2	6×6 mm tactile switches	reset → ext_rst_n; reconfigure → PROGRAMN	2	Robu ₹5 each
JP1	2-pin header + shunt	disables Q1 (DTR auto-reconfigure)	1	Robu ₹5
TP1-TP6	1 mm pads or test loops	3V3, 2V5, 1V1, GND, DONE, INITN	6	—
—	0603 resistor + capacitor assortment kit	avoids buying 100 individual reels	1	Robu ₹500–800 — worth it once, and it covers every passive on this board

3.3 Passives

Ref	Value	Package	Qty	Purpose
R1	10 kΩ	0603	1	ext_rst_n pull-up (the pin has an internal pull-up; this is for noise and SW1)
R2	4.7 kΩ	0603	1	PROGRAMN pull-up to 3V3 (TN-02038 Table 6.2)
R3, R4	4.7 kΩ	0603	2	INITN and DONE pull-ups to 3V3
R5, R6, R7	4.7 kΩ / 1 kΩ / 1 kΩ	0603	3	mode straps: CFG1 = 1 (4.7 k to 3V3), CFG0 = 0, CFG2 = 0 (1 k to GND)
R8	1 kΩ	0603	1	MCLK/CCLK pull-up to 3V3
R9–R12	100 Ω	0603	4	series resistors in the four Master-SPI nets — damping, and they bound any contention with GPIOA[1]/[2]/[4]/[6] (§6.4)

Ref	Value	Package	Qty	Purpose
R13-R15	10 kΩ	0603	3	config flash (U2) /CS, /WP, /HOLD pull-ups to 3V3
R16-R19	10 kΩ	0603	4	application flash (U3) /CS, /WP, /HOLD pull-ups; motor_fault_n pull-up
R21-R24	0 Ω	0603	4	UART routing: R21/R23 = CH340C, R22/R24 = J10 (populate one pair only)
R25, R26	4.7 kΩ / 10 kΩ	0603	2	Q1 gate series resistor and gate pull-down
R27-R30	470 Ω	0603	4	user LED series resistors
R31-R34	1 kΩ	0603	4	indicator drives: R31 = Q4 base, R32 = Q2 base, R33/R34 = series for D1/D6
R36, R37	5.1 kΩ	0603	2	USB-C CC1/CC2 Rd (omit with micro-USB)
R38	12.4 kΩ (E96)	0603	1	1V1 feedback divider, top: $0.8 \times (1 + 12.4/33) = \mathbf{1.1006\ V}$ (was 37.4 kΩ with a 100 kΩ bottom — see R39)
R39	33 kΩ	0603	1	1V1 feedback divider, bottom — measure this rail before fitting U1 (§10 step 2). Corrected from 100 kΩ: the MP1584 datasheet keeps the lower divider resistor below 40 kΩ , because the FB pin's $\approx 20\ \mu\text{A}$ bias current would otherwise push the output outside the FPGA's 1.045–1.155 V window
R40	100 kΩ	0603	1	MP1584 FREQ resistor → 900 kHz (datasheet: $R = 180000/f^{1.1}$ with R in kΩ and f in kHz; its own table gives 100 kΩ = 900 kHz)
R41	100 kΩ	0603	1	MP1584 EN series resistor (with C31 = the $\approx 13\ \text{ms}$ sequencing delay)
R42	1 MΩ	0603	1	Q3 gate to GND — this is what turns the load switch on (with C32 = the 2V5 delay). Revised from 47 kΩ: see R43

Ref	Value	Package	Qty	Purpose
C1	100 nF	0603	1	ext_rst_n RC (with R1)
C2-C7	100 nF	0603	6	1V1 decoupling, one per VCC pin (20, 29, 38, 66, 83, 130)
C8-C11	100 nF	0603	4	2V5 decoupling, one per VCCAUX pin (17, 53, 96, 132)
C12-C20	100 nF	0603	9	3V3 decoupling, one per VCCIO pin (9, 16, 36, 43, 70, 86, 100, 122, 137)
C21	100 μ F / 10 V electrolytic + 100 nF	6.3×5.4 mm / 0603	1+1	1V1 bulk (MP1584 is non-synchronous — needs real bulk)
C22	22 μ F / 10 V electrolytic + 100 nF	5×5 mm / 0603	1+1	2V5 bulk
C23	470 μF / 35 V electrolytic + 100 nF	10×10 mm / 0603	1+1	LM2596 input bulk (datasheet value for a 5–12 V input)
C24	220 μF / 16 V electrolytic + 100 nF	8×7 mm / 0603	1+1	LM2596 output bulk for the 3V3 rail
C25	10 μ F	0805	1	3V3 bulk near banks 0/1/6/7
C26	100 nF	0603	1	Y1 oscillator decoupling
C27	100 nF	0603	1	CH340C decoupling
C28, C29	100 nF	0603	2	flash decoupling (one per device)
C30	10 μ F in + 10 μ F out	0805	2	AMS1117 input/output — it wants $\geq 10 \mu$ F, and ceramics can make it ring; bench check (OQ-B10)
C31	470 nF	0603	1	MP1584 EN delay capacitor (≈ 13 ms with R41)
C32	4.7 μF	0805	1	Q3 gate→ source delay capacitor: $\tau = (R42 \parallel R43) \cdot C32 = 91 \text{ k}\Omega \times 4.7 \mu\text{F} \approx 428 \text{ ms}$, crossing the AO3401 threshold at ≈ 54 ms from 12 V
C33	100 nF	0603	1	MP1584 bootstrap (BST) capacitor

Ref	Value	Package	Qty	Purpose
C36, C37	22 pF	0603	2	CH340G crystal load capacitors (X1 and X0 to GND). Match to the crystal: 22 pF for a CL = 12 pF crystal, 33 pF for the common CL = 20 pF HC49/US type (WCH quotes both values across datasheet revisions). Omit with a CH340C
R43	100 kΩ	0603	1	Q3 gate→source clamp: with R42 it holds Vgs at $-V_{M_IN} \cdot 1M/1.1M \approx -10.9\text{ V}$ at 12 V, inside the AO3401's $\pm 12\text{ V}$ rating (without it the gate would sit at GND and Vgs = -12 V , right on the absolute maximum)
L1	33 μH / 3 A shielded	8×8 mm	1	LM2596 3V3 inductor (datasheet value)
L2	10 μH / 3 A shielded	6×6 mm	1	MP1584 1V1 inductor
FB1	ferrite bead 600 Ω @ 100 MHz	0603	1	VM_IN filtering ahead of the regulators

Cost band (India, one board): U1 ≈ ₹1,650 is three quarters of the total; everything else together is roughly ₹700-1,100 from tier-1 vendors. The module-first build of \$3.5 trades about ₹200 for a much easier assembly.

3.4 Sourcing the FPGA (the one part you cannot substitute)

- The design is an ECP5 design: `rtl/sv16_pll.sv` instantiates `EHXPLLL`, the boot ROM is initialised ECP5 block RAM, `constraints/ecp5_144tqfp.lpf` is a TQFP-144 pin map, and the bitstream comes out of `nextpnr-ecp5`. No other vendor's part fits, and no other ECP5 package fits either.
- Buy it from an Indian storefront of a global distributor:** digikey.in, mouser.in or in.element14.com. They quote in ₹, ship domestically, and clear customs themselves. Lead time is 40+ weeks when the part is not stocked, so **check availability before you commit to the PCB** — this is open item OQ-B13.
- Acceptable substitutions** (same package, pin-compatible, same or better silicon):
- LFE5U-12F-7TG144C** — speed grade 7 instead of 6 (pin-identical, and often the part you can actually find). **Verified on this tree:** `make bitstream CLKSRC=osc FPGA_SPEED=7 BUILD=build/spd7` completes end to end — 9,407 LUT4 (77.5 %), 18/32 sysMEM, **49.53 MHz Fmax** at the 25 MHz constraint (the speed-6 build measures 44.70 MHz in the same configuration), and the resource guard accepts the LFE5U-12F-7TQFP144 part string. Program it with `make prog FPGA_PART=LFE5U-12F`.
- LFE5U-12F-6TG144I** — industrial temperature grade. Drop-in, usually a little dearer.
- Do not** substitute another density (25F/45F) or package (BG256/BG381): the pin map, the PLL reference input and the whole LPF would have to be redone.
- Want ECP5 silicon in your hands before the PCB exists?** A **Colorlight 5A-75B / 5A-75E** LED controller (₹1,500-3,000 via Amazon.in / IndiaMART / AliExpress) contains an **LFE5U-25F-6BG381C** and its JTAG header is publicly reverse-engineered (q3k/chubby75). It is not this board — different package, different pinout, needs its own LPF — but it is a cheap way to run the SV-16 SoC and this toolchain on real silicon while the PCB is being designed.

3.5 Module-first option (recommended for a first build in India)

Hand-soldering a 0.5 mm-pitch TQFP-144 is the hardest part of this board; the second hardest is getting the two switchers right. If you want the first board to work on the first try, replace the power and console sections with ready-made modules on 4-pin headers — the FPGA section and every net name (3V3 / 2V5 / 1V1 / VM_IN, uart_tx / uart_rx) stay exactly the same:

Instead of	Fit	₹	Why it is easier
U5 + L1 + D10 + C23 + C24 (3V3)	an LM2596 buck module (Robu ₹43–50) with its output set to 3.3 V	45	no inductor/diode/feedback to get wrong; the pot sets the voltage
U6 + L2 + D11 + R38/R39/R40/C31 + C21 (1V1)	a Mini360 / MP1584 module (Robu ₹105), output trimmed to 1.10 V, measured with a meter	105	same reason; the divider risk disappears
U7 + Q3 + C30/C32 (2V5)	a second small buck module set to 2.5 V, or an AMS1117-2.5 breakout	50–105	
U4 + U8 + J8 (USB-UART)	a CH340C + USB-C serial breakout (Robu ₹149) — 4-pin header, cross TX/RX	149	no SOP-16, no USB routing, no ESD array
J2 + J1 (JTAG)	keep J1 (2.54 mm) — modules cannot replace this	10	

Two caveats: modules make the \$4.3 power-up order less deterministic (many MP1584 modules expose an EN pad, so you can still add the RC delay — otherwise keep the PROGRAMN-hold fallback), and they are bulky: reserve ~35 × 45 mm for the module area and mount them on headers rather than soldering them down, so you can swap one out.

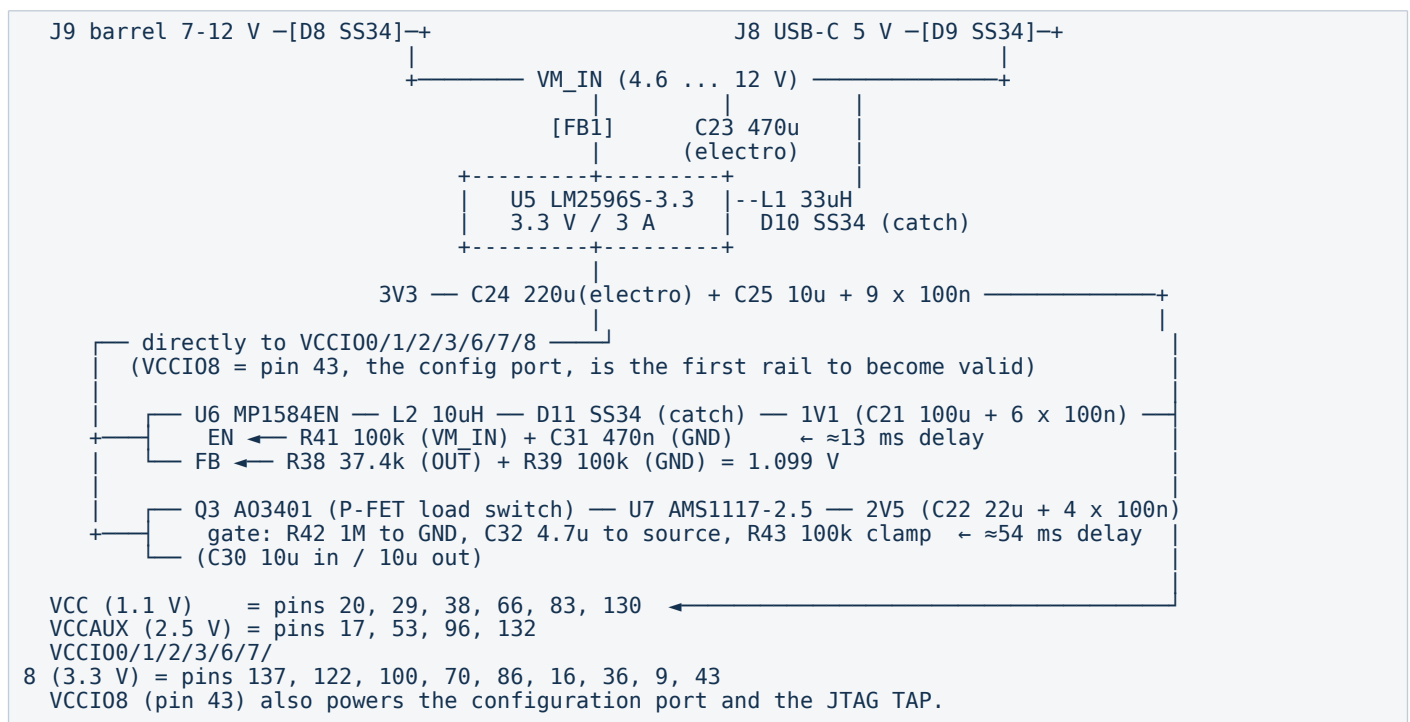
3.6 What changed from revision 1.0, and why each replacement is equivalent

Rev 1.0 part	Now	Why it is an equivalent (not just "something similar")
U5 AP63203 (synch. buck, TSOT-26)	LM2596S-3.3 (TO-263-5)	Same job: 3.3 V rail ≥ 2 A from a 5–12 V input. Slower (150 kHz vs 1.1 MHz) and non-synchronous, so it needs an external Schottky (D10) and much more bulk capacitance (C23/C24) — but 3 A rated and it is on the shelf in India for ₹51. Neutral for the FPGA: the rail is 3.3 V either way
U6 TPS62823 (synch. buck, 2.4–5.5 V in)	MP1584EN (SOIC-8E)	Same job: 1.1 V core rail, 3 A. Difference: the MP1584 needs ≥ 4.5 V input , so it runs from VM_IN instead of from 3V3 — that is what forces the explicit sequencing in §4.3. 0.8 V reference (vs the TPS62823's lower one) sets 1.1 V with a divider, still inside the FPGA's 1.045–1.155 V window with 5 % margin
U7 LP5907-2.5 (LDO, SOT-23-5)	AMS1117-2.5 (SOT-223)	Same job: 2.5 V for VCCAUX at 16 mA. Bigger, noisier and it needs ≥ 10 μ F output — irrelevant for a 16 mA load of reference input buffers
Y1 ASEMB-25.000 (XO, 3.2×2.5)	YIC OSC25M-3.3I/S3-25T	Same job: 25.000 MHz, 3.3 V CMOS, ± 25 ppm (better than the ± 50 ppm the ± 2 % UART tolerance needs). Any 25 MHz 3.3 V active oscillator in 5×7 mm works
R38–R42, C30–C33, D10, D11, Q3	new	Sequencing and the passives the two new bucks require

Rev 1.0 part	Now	Why it is an equivalent (not just "something similar")
passives 0402	0603	Same values, hand-solderable; 0402 stays only if you shrink the board
"no electrolytics anywhere"	470 μ F + 220 μ F + 100 μ F + 22 μ F electrolytics	Consequence of the two non-synchronous bucks
L1 4.7 μ H, L2 2.2 μ H	33 μH (LM2596) and 10 μH (MP1584)	Inductor values are set by the switcher, not the load

4. Power architecture

4.1 Rail generation



4.2 Rails and current budget

Rail	Pins (count)	Nominal	Allowed range (datasheet)	Consumers	Static budget
1V1 (VCC)	20, 29, 38, 66, 83, 130 (6)	1.10 V — set by R38/R39	1.045-1.155 V (Table 3.2); abs max 1.32 V (Table 3.1)	FPGA core	77 mA typical (Table 3.8, T _J = 85 °C) + switching current; U6 rated 3 A
2V5 (VCCAUX)	17, 53, 96, 132 (4)	2.50 V	2.375-2.625 V; abs max 2.75 V	differential/referenced input buffers	16 mA typical (Table 3.8) + 5 mA LDO quiescent; U7 rated 1 A
3V3 (VCCIO + I/O)	9, 16, 36, 43, 70, 86, 100, 122, 137 (9)	3.30 V	3.135-3.465 V for LVCMOS33 (Table 3.11); abs max 3.63 V	6 banks + 2 flashes + oscillator + LEDs + CH340C + the 2V5 LDO's 5 mA	6 x 0.5 mA typical (Table 3.8) + loads + U7 quiescent; U5 rated 3 A

Rail	Pins (count)	Nominal	Allowed range (datasheet)	Consumers	Static budget
VM_IN	J9 / J8	7–12 V (barrel, recommended) or 5 V (USB)	U5 input 4.5–30 V, U6 input 4.5–28 V — see the USB note below	U5, U6, the 5 V pins of J3/J4, J6 motor passthrough	external supply $\geq$ 1 A

USB-C power is at the input-voltage edge. 5 V USB minus the SS34 drop (~ 0.4 V) leaves ≈ 4.6 V at VM_IN, and both bucks want ≥ 4.5 V. That works for bench use at light load, but the margin disappears as soon as the motor header is loaded — **use the barrel jack at 7–12 V for anything real**, or replace D9 with a P-MOSFET ideal-diode. Open item OQ-B12.

Timing margin is not the same thing as power margin. The static numbers above are from the datasheet; this repository has **no** dynamic power estimate for the SoC (the open toolchain has no power analyser, and `nextpnr` emits none). The regulators are therefore specified with a large margin, and "measure the real 1V1 current during bring-up" is an explicit bring-up step (§10) and an open item (§13.2 OQ-B1). Nothing on the board draws more than 3 A on any rail even in the worst case, and the 1V1 buck is current-limited, so an over-estimate cannot damage the device.

4.3 Power-up order, POR and the rules the board must not break

Rule	Requirement	How this board satisfies it
Recommended levels	VCC 1.045–1.155 V, VCCAUX 2.375–2.625 V, VCCIO 1.14–3.465 V	fixed-value regulators: 1.10 V / 2.50 V / 3.30 V
Ramp rate	all supplies 0.01–10 V/ms; VCCAUX $\leq$ 30 mV/ μ s	LM2596 soft-start $\approx$ 1 V/ms; MP1584 $\approx$ 0.5 V/ms; AMS1117 $\approx$ 0.1 V/ms — all inside the window (verify on the bench, §10 step 3)
POR release	only when VCC, VCCAUX and VCCIO8 are all above their trip points (0.90–1.00 V, 2.00–2.20 V, 0.95–1.06 V)	all three rails are generated and all three are monitored by the device itself; no board-side reset generator is needed
Master-SPI order (§3.5)	ramp VCCIO8 above the flash V_{IH} (0.7×3.3 V $\approx$ 2.31 V) before VCC or VCCAUX reach their POR trip (0.90–1.00 V and 2.00–2.20 V), or hold PROGRAMN/INITN low until VCCIO8 is valid	explicitly sequenced — see the timeline below. Because the India-sourced bucks need ≥ 4.5 V input, VCC and VCCAUX are no longer derived from 3V3, so the order is created with two RC networks instead of by construction
Fallback if the bench disagrees	the datasheet's second option	drive PROGRAMN low from the host (J1.4 / the CH340C DTR path through Q1) and release it after 3V3 is up; this is a firmware/host action, not a board change
VCCIO8 ramp-down	the device has no ramp-down detection on VCCIO8; a brown-out that only drops VCCIO8 while VCC stays valid can leave the part in an undefined state	the 3V3 rail is the master: if the supply collapses, VCC collapses with it (all rails come from the same VM_IN), so the device sees a clean power cycle. Do not add a separate 3V3-only cutoff switch
Sequencing on power-down	no requirement	bulk capacitance is sized so all three rails fall together; the 3V3 rail has the largest load (C23/C24/C25) and therefore falls last, which is what the datasheet wants
JTAG VREF	the 1×10 Versa-style header has no VREF pin, the 2×5 header does	J1.1 and J2.1 both sit on 3V3; adapters that measure VREF will read 3.3 V, and adapters that level-shift from VREF will drive 3.3 V logic correctly

Power-up timeline (typ, 12 V input). $t = 0$ is VM_IN applied.

Time	Event	Consequence
≈0.3 ms	3V3 (U5, LM2596 soft-start) crosses 2.31 V	VCCIO8 is now above the config flash's V_{IH} — the ordering requirement is met with two orders of magnitude of margin
≈13 ms	U6 starts (EN released by R41/C31; threshold ≈1.2 V at 5 V in: $t = -RC \cdot \ln(1 - 1.2/V_{in})$, $RC = 47$ ms)	VCC begins to rise after VCCIO8 is fully valid
≈14 ms	1V1 crosses 0.90 V	VCC POR trip, comfortably after VCCIO8
≈54 ms	Q3 turns on (gate falls with $\tau = (R42 \parallel R43) \cdot C32 \approx 428$ ms, crossing the −1.3 V threshold) → 2V5 rises	VCCAUX POR trip, last of the three
≈60 ms	POR releases (VCC, VCCAUX and VCCIO8 all above trip)	the device starts configuration from U2

4.4 Decoupling

Rail	Bulk	Per-pin ceramic	Placement rule
1V1	C21 100 μ F electrolytic	C2–C7 100 nF (6, one per VCC pin)	100 nF within 3 mm of each pin, via to the ground plane directly beside the pad
2V5	C22 22 μ F electrolytic	C8–C11 100 nF (4, one per VCCAUX pin)	same
3V3	C23 470 μ F (input) + C24 220 μ F (output) electrolytic + C25 10 μ F ceramic	C12–C20 100 nF (9, one per VCCIO pin)	same; the 100 nF on VCCIO8 (pin 43) is the most important one on the board — it also decouples the configuration port
Input	—	100 nF per regulator pin + FB1	at the input diodes

The electrolytics are the price of using two non-synchronous bucks; keep them close to their regulator rather than near the FPGA, and keep the switching loops small (§11).

5. Clocking

```
Y1 (25.000 MHz CMOS X0, 3.3 V) — CLK25 → U1 pin 133 clk_25m
| power: 3V3 + C26 100 nF
| output enable (pin 1): tied to 3V3 (always on)
```

Item	Value	Source
Frequency	25.000 MHz	constraints/ ecp5_144tqfp.lpf (clk_25m), docs/RESET_AND_CLOCK.md
Stability	±25 ppm (the India-available YIC part) — far better than needed: the console runs 115200 8N1 (≈ ±2 % tolerance) and the tests are cycle-based	this design

Item	Value	Source
Sourcing	it must be an active oscillator, not a crystal. ECP5 has no on-chip oscillator and the design expects a driven <code>clk_25m</code> , so a ₹39 passive 25 MHz crystal (the "YSX321SL" type that Indian shops list) will not work. Buy the YIC part from digikey.in (₹111) or any 25 MHz 3.3 V active oscillator in 5×7 mm / 3.2×2.5 mm from Amazon.in / eBay.in / Sunrom	
PLL configuration	CLKI_DIV 2, CLKFB_DIV 3, CLKOP_DIV 16 → CLKOP = 37.5 MHz, VCO = 600 MHz (inside the 400–800 MHz range)	<code>rtl/sv16_pll.sv</code> , docs/ARCHITECTURE_DECISIONS.md ADR-024
Achieved timing	45.45 MHz Fmax at the 37.5 MHz constraint (<code>build/pll/sv16_nextpnr.log</code>), 44.70 MHz in the <code>CLKSRC=osc</code> 25 MHz build	docs/SYNTHESIS_AND_DEPLOYMENT.md
Second clock	none. No DDR, no SERDES (the ECP5 144-TQFP offering has 0 SerDes channels), no second oscillator input — the design has a single clock domain plus the hardened blocks	datasheet Table 1.1 / §4.3.2
JTAG clock	$TCK \leq 25$ MHz (<code>TAP_SCAN_CLOCK</code> of <code>TCK</code> is (25.0e6, B0TH) in the BSDL model)	FPGA-MD-02097
Configuration clock	25 MHz pin 54 output in Master SPI; the device starts at ≈ 2.4 MHz and may step up per the bitstream's <code>MCCLK_FREQ</code> . A 293 KB compressed image therefore takes ≈ 1 s to load	TN-02039 §6.1, estimate

6. FPGA pin map and the board nets

`board/TQFP144_PINOUT.md` is the complete, generated 144-pin table (device pad, bank, rail, design signal, drive strength, pull mode and board destination). This section is the human-readable summary.

6.1 Signals used by the current design (52 of 98 I/O pins)

Function	Bank	Pins (TQFP144)	Drive / pull (from the LPF)	Board destination
25 MHz clock	0	133	4 mA	Y1.3
Reset button input	0	134	4 mA / pull-up	SW1 + R1 + C1
Console UART	3	73 (rx), 74 (tx)	4 mA	CH340C (U4) / J10
Application flash (bank-1 SPI)	1	110 sck, 111 cs_n, 112 mosi, 113 miso	8/8/8/4 mA	U3 (application flash)
SPI0 expansion	1	114 sck, 115 cs_n, 116 mosi, 117 miso	8/8/8/4 mA	J3 (EXP-A)
Status LEDs (mirror of GPIOA[3:0])	8	39, 40, 41, 44	8 mA	D2–D5 + R27–R30
GPIOA[0:7]	8	45–52	8 mA	J5 EXP-C + config-flash nets for [1],[2],[4],[6] (§6.4)
GPIOA[8:15]	2	97, 98, 99, 104, 105, 106, 107, 108	8 mA	J3 EXP-A

Function	Bank	Pins (TQFP144)	Drive / pull (from the LPF)	Board destination
GPIOB[0:7]	0	135, 136, 139, 140, 141, 142, 143, 128	8 mA	J4 EXP-B
GPIOB[8:11]	1	124, 125, 126, 127	8 mA	J4 EXP-B
GPIOB[12:15]	7	1, 2, 3, 4	8 mA	J4 EXP-B
PWM / motor	2	88 pwm (16 mA), 89 dir1, 102 dir2, 103 fault_n (pull-up)	see left	J6 motor header

LEDs are not independent software GPIO: `rtl/sv16_top.sv` does `assign led = ~gpio_a_out[3:0]`, so the LED pins always mirror GPIOA[3:0] and light when the bit is **0** (active low). GPIOA[3:0] are also on J5 pins 4/5/6/7 — the header copy and the LED copy are separate FPGA pins, so a lever on J5 does not have to disturb the LED indication, but both reflect the same register bit.

6.2 Spare I/O (46 pins)

Bank	Free pins	Where brought
1	118, 119, 120, 121	J7
2	90, 91, 92, 93, 94, 95	J7
3	67, 68, 69, 71, 72, 76, 77, 78, 79, 80, 81, 82, 84	J7
6	18, 19, 22, 23, 24, 25, 26, 27, 28, 30, 31, 33, 34, 35, 37	J7
7	5, 6, 7, 10, 11, 12, 13, 14	J7

Banks 6 and 7 are unused by the current bitstream, so adding a peripheral to them costs nothing in LUTs today. Two notes for whoever uses them: bank 3 pin 79 (PR35B, VREF1_3), bank 6 pin 19 (PL35B, VREF1_6) and bank 7 pin 11 (PL14C, VREF1_7) are VREF inputs, and bank 2 pin 88 is the 16 mA PWM output — do not stack heavy loads on it (it is already the motor PWM).

6.3 Active-low and direction conventions

Signal	Direction	Idle / reset state	Consequence
ext_rst_n	input	high (pull-up), press = low	SW1 resets the SoC; the startup FSM then runs the boot sequence
led[3:0]	output	LEDs off (GPIOA[3:0] = 1 after reset)	active low
flash_cs_n, spi0_cs_n	output	high = deselected	active low
motor_fault_n	input	pulled up = "no fault"	active low; the motor block at 0xF030 reports it (docs/PERIPHERALS.md)
uart_rx	input	pulled up; must not be held low at power-up	the startup FSM forces the monitor after 4096 clocks of a held-low RX (RX_HOLD_CYCLES) — deliberate, and the reason the CH340C's TXD idles high
all GPIO	bidirectional	dir_reg = 0 on reset ⇒ inputs (high-Z)	this is what makes §6.4 safe in software

6.4 The one hard constraint on this board: the Master-SPI pads are also GPIOA pins

The design constrains gpio_a[0:7] to bank-8 pins 45–52. Four of those pads are the device's Master-SPI configuration port:

TQFP pin	Pad	sysCONFIG role	Design signal	Board net
46	PB11A	D1/MISO — input from the config flash	gpio_a[1]	J5.2 and U2.2 (flash DO) via R11 100 Ω
47	PB11B	D0/MOSI — output to the config flash	gpio_a[2]	J5.3 and U2.5 (flash DI) via R10 100 Ω
49	PB15A	HOLDN/DI/BUSY/CSSPIN — flash chip select	gpio_a[4]	J5.5 and U2.1 (/CS) via R12 100 Ω
51	PB15B	DOUT/CS0N — unused in a single-FPGA chain	gpio_a[6]	J5.7 only (no flash connection)
39, 40, 41, 44, 45, 48, 50, 52	PB6A, PB4A, PB6B, PB4B, PB9A, PB13A, PB13B, PB18A	parallel / slave-SPI pads, not claimed by Master SPI	led[0:3], gpio_a[0], [3], [5], [7]	headers / LEDs, free

Why this is nevertheless safe, and what must be true for it to stay safe:

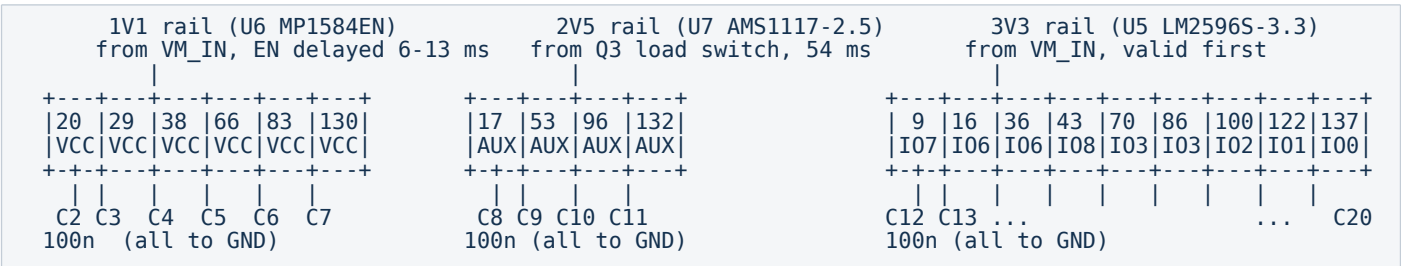
- During configuration the configuration logic drives pins 54, 47, 46 and 49 and the fabric is held in reset, so the user I/O on those pads cannot fight it.
- After configuration, the pads are released to the fabric **because the design does not set the PERSISTENT attribute** on the spi port. This is what "recovering the configuration port pins for use as general purpose I/O" means in TN-02039 §4.6.10. Keeping it that way is deliberate.
- The **board uses a separate flash for firmware** (U3 on bank-1 pins 110–113). The bitstream flash (U2) is only touched by the configuration logic and by JTAG, never by the soft-core.
- The four shared nets carry 100 Ω series resistors (R9–R12). If firmware ever drives gpio_a[1], [2], [4] or [6] as an output while U2 is driving DO, the series resistor bounds the current instead of letting two drivers fight. Software must still treat those four bits as **inputs** (they are inputs after reset, and nothing in the ROM monitor or the bootloader ever changes that).
- Consequence for a user: J5 pins 2, 3, 5 and 7 are not free general-purpose pins on this board — they are the configuration flash's DO, DI, /CS and the unused DOUT line. Use a spare pin from §6.2 if you need a clean signal, or apply variant V1 (§14) which remaps those four GPIO bits.

Everything else about the pin map is uncomplicated: no pin used by the design is shared with the JTAG TAP, and no pin used by the design is one of the dedicated sysCONFIG pins (PROGRAMN 57, INITN 55, DONE 56, CCLK 54, CFG_0 62, CFG_1 59, CFG_2 58, TCK 63, TMS 64, TDI 61, TDO 60).

7. Connection circuit diagram

Sheets 1–5 below are the connection diagram. Net labels are the LPF signal names, so any net can be traced back to a line in constraints/ ecp5_144tqfp.lpf and to a row of board/TQFP144_PINOUT.md. Pin numbers for U1 are TQFP-144 physical pins; pin numbers for the flashes are SOIC-8 as defined by the W25Q datasheet; the CH340C is drawn by pin name (the vendor's SOP-16 numbering applies).

7.1 Sheet 1 — FPGA supply and ground



```

+ C21 100u (electro)          + C22 22u (electro)          + C24 220u + C25 10u

U5 (3V3): VM_IN → [C23 470u electro + 100n] → IN ; OUT → L1 33uH → 3V3 ; D10 SS34 from OUT to GND(anode)
OUT → C24 220u → 3V3 ; ON/OFF pin → GND (always enabled) ; GND → plane
FB (pin 4) → 3V3 : the fixed-output version senses its own output here, so this pin
is NOT left floating (the TI datasheet's force-on/force-off tests are written as
"feedback pin removed from output")
U6 (1V1): VM_IN → [100n] → IN ; EN ← VM_IN via R41 100k with C31 470n to GND (13 ms at 5 V in,
~6 ms at 12 V in: EN rising threshold 1.5 V, falling 1.2 V, tau = 47 ms)
SW → L2 10uH → 1V1 ; D11 SS34 catch to GND ; BST → C33 100n → SW
FB ← R38 12.4k from 1V1, R39 33k to GND (= 1.1006 V) ; C21 100u on the 1V1 rail
FREQ → R40 100k → GND (= 900 kHz) ; GND (pin 5) and the exposed pad → plane
COMP (pin 3): needs an RC compensation network to GND – values from the MP1584
datasheet's typical application for 1.1 V out at 900 kHz (open item OQ-B10)
Q3/
U7 (2V5): VM_IN → Q3 A03401 (S) → (D) → U7 AMS1117-2.5 IN → OUT = 2V5 → C22 22u
(the AMS1117 needs >= 3.8 V in for a 2.5 V out, so this section runs from VM_IN,
not from 3V3)
Q3 gate: R42 1M to GND + C32 4.7uF to source + R43 100k source-to-gate (Vgs clamp);
the gate starts at source potential (off) and falls with tau = 91k x 4.7uF = 428 ms,
reaching the threshold at ~54 ms from 12 V
U7: C30 10u in, 10u out ; GND → plane

Ground: 8 (VSSI07) 15 (VSSI06) 21 (VSS) 32 (VSSI06) 42 (VSSI08) 65 (VSS) 75 (VSSI03)
85 (VSS) 87 (VSSI03) 101 (VSSI02) 123 (VSSI01) 129 (VSSI00) 131 (VSS) 138 (VSSI00)
-> every pin to the GND plane with its own via. Pins 109 and 144 are NC.

```

7.2 Sheet 2 — Configuration mode, JTAG, reset and status

```

U1.57 PROGRAMN --+--[R2 4.7k]-- 3V3
+-- SW2 --- GND (press = reconfigure)
+-- J1.4 (header PROGRAMn)
+-- Q1 drain ; Q1 source -- GND ; Q1 gate --[R25 4.7k]-- JP1 -- U4 DTR#
Q1 gate --[R26 10k]-- GND
(auto-reconfigure from the host, disable by removing JP1)

U1.55 INITN -----+--[R3 4.7k]-- 3V3
+-- J1.10 (header INITn)
+--[R31 1k]-- Q4 base Q4 = BC807 (PNP, optional indicator)
emitter -- 3V3 ; collector -- D6 --[R34 1k]-- GND
(D6 red lights when INITN is low = SRAM clear / config error)

U1.56 DONE -----+--[R4 4.7k]-- 3V3
+-- J1.9 (header DONE)
+--[R32 10k]-- Q2 base Q2 = BC817 (NPN)
Q2 emitter -- GND ; Q2 collector -- D1 --[R33 1k]-- 3V3
(D1 green lights when DONE is high = configured /

in user mode)

Mode straps (Master SPI, CFG[2:0] = 010, latched on the rising edge of INITN):
U1.62 CFG_0 --[R6 1k]-- GND
U1.59 CFG_1 --[R5 4.7k]-- 3V3
U1.58 CFG_2 --[R7 1k]-- GND

JTAG TAP (dedicated pins, VCCI08 domain):
U1.63 TCK -- J1.8 -- J2.5
U1.64 TMS -- J1.6 -- J2.2 J1: 1=3V3(PWR) 2=TDO 3=TDI 4=PROGRAMn 5=NC
U1.61 TDI -- J1.3 -- J2.3 6=TMS 7=GND 8=TCK 9=DONE 10=INITn
U1.60 TDO -- J1.2 -- J2.4 J2: 1=3V3(VREF) 2=TMS 3=TDI 4=TDO 5=TCK
6=GND 7..9=NC 10=GND

JTAG works with a blank or corrupt configuration flash -> this is the recovery path.

```

7.3 Sheet 3 — The two SPI flashes

```

U2 = W25Q64JVSSIQ – FPGA bitstream flash (Master SPI, driven by the configuration logic)
-----
U1.54 CCLK --[R9 100R]-----> U2.6 CLK ; U1.54 --[R8 1k]-- 3V3
U1.47 PB11B (MOSI pad) --[R10 100R]--> U2.5 DI ; also J5.3 as gpio_a[2] *
U1.46 PB11A (MISO pad) --[R11 100R]--> U2.2 DO ; also J5.2 as gpio_a[1] *
U1.49 CSSPIN pad --[R12 100R]--> U2.1 /CS ; also J5.5 as gpio_a[4] *
U2.1 --[R13 10k]-- 3V3 (deselect when idle)
U2.3 /WP --[R14 10k]-- 3V3
U2.7 /HOLD --[R15 10k]-- 3V3 (I02/I03 not used: no quad on TQFP144)
U2.8 VCC = 3V3 + C28 100n ; U2.4 GND
* the 100R series resistors bound any contention between the pads and the flash; firmware keeps
gpio_a[1]/[2]/[4]/[6] as inputs (see §6.4)

U3 = W25Q64JVSSIQ – SV-16 application flash (firmware slots, driven by the soft-core)
-----
U1.110 flash_sck -----> U3.6 CLK

```

```

U1.111 flash_cs_n -----> U3.1 /CS    ; U3.1 --[R16 10k]-- 3V3
U1.112 flash_mosi -----> U3.5 DI
U1.113 flash_miso <----- U3.2 DO
                        U3.3 /WP --[R17 10k]-- 3V3
                        U3.7 /HOLD --[R18 10k]-- 3V3
                        U3.8 VCC = 3V3 + C29 100n ; U3.4 GND

```

Why two devices: the configuration port is not memory-mapped, so the soft-core cannot reach U2. U3 is an ordinary SPI device on user I/O, which is what sv16_flash_ctrl (MMIO 0xF0A0) drives and what the bootloader reads firmware images from. See §6.4 for the full argument.

7.4 Sheet 4 — Clock, reset and console

```

Y1 25.000 MHz ACTIVE X0 (3.3 V)      U4 CH340G (SOP-16, 3.3 V, pin numbers below)
-----
Y1.3 OUT ----> U1.133 clk_25m        U4.16 VCC -- 3V3 + C27 100n
Y1.1 EN  -- 3V3                      U4.4 V3  -- 3V3 + 100n **
Y1.4 VCC -- 3V3 + C26 100n          U4.1 GND -- GND
Y1.2 GND -- GND                    U4.5 UD+ <--> J8 D+ (via USBLC6 D+ channel)
                                U4.6 UD- <--> J8 D- (via USBLC6 D- channel)
** a passive crystal (HC49/US, 2 pins) CANNOT drive this net. The ECP5 has no on-chip crystal
   oscillator, so Y1 must be a 4-pin ACTIVE oscillator with its own VCC. See OQ-B14.

```

CH340G clock (only for the G version; a CH340C has this inside and needs none of it)

```

U4.7 XI --- Y2 12 MHz --- U4.8 X0
(or, instead of Y2/C36/C37: an active 12 MHz oscillator into XI, X0 left open)
      |           |
      C36 22pF    C37 22pF
      |           |
      GND         GND
(keep XI/X0 short and symmetrical; do not route them near the USB pair or the switchers)

```

```

U4.2 TXD --[R21 0R]-- U1.73 uart_rx (idles high)
U4.3 RXD --[R23 0R]-- U1.74 uart_tx
U4.13 DTR# -- JP1 -- Q1 gate (optional auto-reconfigure, §7.2)
U4.15 R232 -- NC (internal pull-down; leave open)
U4.9 CTS# / 10 DSR# / 11 RI# / 12 DCD# / 14 RTS# -- NC

```

SW1 reset

```

U1.134 ext_rst_n --- SW1 -- GND
                  +---[R1 10k]-- 3V3
                  +--- C1 100n -- GND
U1.134 has PULLMODE=UP in the LPF; R1/C1 are for noise immunity and contact bounce,
not for correct reset behaviour.

```

J10 alternative console (populate R22/R24 instead of R21/R23)

```

J10.1 3V3 | J10.2 UART_TX(F) --[R24 0R]-- U1.74 | J10.3 UART_RX(F) --[R22 0R]-- U1.73 | J10.4 GND
** 3.3 V-only board: a 5 V USB-UART adapter must not be used unless a level shifter is fitted
   (BOM "optional" row) or the adapter's I/O is 3.3 V tolerant.

```

7.5 Sheet 5 — LEDs, expansion and motor

Status LEDs (mirror of GPIOA[3:0], active low -- see §6.1)

```

U1.39 led[0] --[R27 470R]-- D2 -- 3V3      (D2 on when GPIOA[0] = 0)
U1.40 led[1] --[R28 470R]-- D3 -- 3V3
U1.41 led[2] --[R29 470R]-- D4 -- 3V3
U1.44 led[3] --[R30 470R]-- D5 -- 3V3

```

Expansion headers (every signal is 3.3 V LVCMOS, 8 mA drive as constrained in the LPF)

```

J5 EXP-C : 1 A0  2 A1*  3 A2*  4 A3   5 A4*  6 A5   7 A6*  8 A7   9 GND 10 3V3
J3 EXP-A : 1 3V3  2 GND  3 A8   4 A9   5 A10  6 A11  7 A12  8 A13  9 A14 10 A15
          11 spi0_cs_n 12 spi0_sck 13 spi0_mosi 14 spi0_miso
          15 GND 16 GND 17 5V 18 5V 19 GND 20 GND
J4 EXP-B : 1 3V3  2 GND  3 B0   4 B1   5 B2   6 B3   7 B4   8 B5   9 B6  10 B7
          11 B8  12 B9  13 B10 14 B11 15 B12 16 B13 17 B14 18 B15 19 GND 20 5V
J7 spare : 2x25, 46 spare I/O (bank 1 : 118-121, bank 2 : 90-95, bank 3 : 67,68,69,71,72,
          76-82,84, bank 6 : 18,19,22-28,30,31,33-35,37, bank 7 : 5-7,10-14) + 4 GND.
          Footprint only, not fitted.

```

* A1/A2/A4/A6 are the shared configuration-flash nets of §6.4; A3/A5/A7 are free.

The 5 V pins are USB VBUS only (the 3V3 rail comes from VM_IN, §4.1).

Motor connector (keyed 1x6)

```

J6.1 VM <-- VM_IN through the keyed header (fused on the driver board, not on this board)
J6.2 GND -- GND
J6.3 PWM <----- U1.88 pwm_out (16 mA drive, PWM block at MMIO 0xF030)
J6.4 DIR1 <----- U1.89 motor_dir1 (mirror of GPIOA[4])
J6.5 DIR2 <----- U1.102 motor_dir2 (mirror of GPIOA[5])
J6.6 FAULTn <-----> U1.103 motor_fault_n (input, PULLMODE=UP + R19 10k pull-up to 3V3)

```

8. Connector pinouts and how to use them

Connector	Type	Pinout	Use
J1	1×10, 2.54 mm	1 3V3 · 2 TDO · 3 TDI · 4 PROGRAMn · 5 NC · 6 TMS · 7 GND · 8 TCK · 9 DONE · 10 INITn	Versa-style programming header, works with openFPGALoader and any FT232H/FT2232H adapter
J2	2×5, 1.27 mm keyed	1 3V3 (VREF) · 2 TMS · 3 TDI · 4 TDO · 5 TCK · 6 GND · 7-9 NC · 10 GND	Digilent/HS2-style probes; wired in parallel with J1 — use one at a time
J3	2×10, 2.54 mm	see §7.5	SPI0 + GPIOA[15:8]; SPI0 is the SoC's second SPI (0xF050), ideal for an SD card or radio module
J4	2×10, 2.54 mm	see §7.5	all 16 GPIOB pins (0xF070)
J5	1×10, 2.54 mm	see §7.5	GPIOA[7:0]; pins 2/3/5/7 share the configuration flash (§6.4)
J6	1×6 keyed	see §7.5	motor driver interface
J7	2×25, 2.54 mm	46 spare I/O + 4 GND	expansion only; not fitted
J8	USB-C receptacle	VBUS, D+, D-, GND, CC1/CC2 = 5.1 kΩ Rd	power + console on one cable
J9	5.5/2.1 mm barrel	centre positive, 7-12 V (5 V USB works only at light load — OQ-B12)	bench supply, powers the motor connector too
J10	1×4, 2.54 mm	3V3 · UART_TX(F) · UART_RX(F) · GND	console without fitting the CH340C (fit R22/R24, remove R21/R23)

Wiring notes that prevent the two most common support cases:

- **A 5 V USB-UART adapter on J10 will damage the FPGA pins** (they are LVCMOS33 with no 5 V tolerance documented for this configuration). Use 3.3 V levels, or fit the optional level shifter.
- **Two JTAG adapters on J1 and J2 at the same time** will fight. Populate one connector if you are unsure; they are on the same four nets.
- The **console is the only update path in the field**: bitstreams go through JTAG (or through U2), while firmware images go through the UART into U3. That asymmetry is inherent to soft-core + configuration flash and is worth putting on the silkscreen.

9. What lives in which memory, and how each part gets programmed

Storage	Device	Content	Written by	Survives power cycle
Configuration SRAM (inside U1)	—	the FPGA bitstream	configuration logic from U2, or JTAG directly	no
U2 bitstream flash	W25Q64JV	build/p11/sv16_top.bit (293 KB) at offset 0	openFPGALoader -f through the FPGA's MSPI port, or a bench SPI programmer (SOIC-8 clip)	yes
U3 application flash	W25Q64JV (W25Q40-class is enough, see below)	firmware images: slot A at 0x000000, slot B at 0x008000, 32 KB each, 32-byte header, slot record at bytes 0x18/0x19	the SV-16 ROM monitor over the UART (make upload, make upload-slot), or an external SPI programmer	yes
Boot ROM	hardened in the FPGA (initialised BRAM, .ORG 0xE000)	the ROM monitor (790 lines of SV-16 assembly)	bitstream build (make rom)	n/a (part of U2)
SRAM	FPGA block RAM	32 KB at 0x0000-0x3FFF — applications at run time, loader while booting	boot loader / firmware	no

docs/

BOOT_AND_PROGRAMMING.md documents the image format, the A/B slot policy (PENDING → TRIED → GOOD or BAD), and the monitor's command set; this board only supplies the two physical flash devices it assumes. The typical application flash in that document is a **512 KB W25Q40-class part**, so U3 can be downsized to W25Q40CLSNIQ (or W25Q16JV) without changing a line of firmware — the design only needs the standard SPI-NOR command set (0x03/0x0B read, 0x02/0x06/0x05/0x20/

0x08 program/erase/status) that the whole W25Q family shares. U2 must be at least as large as the bitstream: 1 MB is comfortable, 8 MB is what the BOM lists because the part is the same and cheaper in volume.

9.1 Programming flows on this board

```
# 1. Volatile configuration (fastest loop while developing the FPGA side)
make bitstream          # -> build/sv16_top.bit (J1/J2 + any FT232H/FT2232H adapter)
make prog CABLE=ft232   # openFPGALoader --fpga-part LFE5U-12F build/sv16_top.bit

# 2. Persistent bitstream (survives power cycles, loads from U2 in ~1 s at the default MCLK)
make prog-flash CABLE=ft232 # openFPGALoader -f: writes U2 through the FPGA's MSPI port

# 3. Firmware: single-image upload through the monitor (no rollback)
make upload PORT=/dev/ttyUSB0 IMG=build/fw/app_img.hex

# 4. Firmware: A/B field update (recommended -- a bad image cannot brick the board)
make slot-image SLOT=1 && make upload-slot SLOT=1 PORT=/dev/ttyUSB0
make mon-boot          # boot the PENDING slot; loader marks it TRIED
make commit            # once the new image behaves: record GOOD, retire the other slot
```

The JTAG headers stay useful even with a blank U2: openFPGALoader --detect should identify the device (the ECP5 12F IDCODE is device specific — record what your part reports; the 25F model in the BSDL FPGA-MD-02097 reads 0x41111043), then flow 1 or 2 recovers the board.

10. Bring-up and acceptance test plan

#	Step	Pass criterion	Tool
1	Resistance check before power	> 100 Ω from each rail (TP1/TP2/TP3) to GND; no short between rails	DMM

#	Step	Pass criterion	Tool
2	Power up on the barrel jack (7–12 V) with U1 not fitted yet	TP1 = 3.135–3.465 V, TP2 = 2.375–2.625 V, TP3 = 1.045–1.155 V — the 1V1 rail is now set by the R38/R39 divider, so verify and trim it here , before the FPGA is on the board and can be damaged by an out-of-range core rail	DMM
2b	With all three rails in spec, fit U1 (and only then, the flashes)	—	—
3	Ramp order (scope, two channels)	3V3 rises first and passes 2.31 V before either of the other rails moves at all; 1V1 starts ≈13 ms later, 2V5 ≈50 ms later; no rail slower than 0.01 V/ms; VCCAUX slew ≤ 30 mV/μs	scope, 20 ms/div
4	JTAG detection	<code>openFPGALoader --detect</code> returns the ECP5 IDCODE, no cable/chain errors	J1/J2 + adapter
5	Volatile configuration	DONE (TP5) goes high, D1 lights, INITN stays high	<code>make prog</code>
6	Console	115200 8N1 shows the monitor banner; ? lists commands	<code>make mon-term PORT=/dev/ttyUSB0</code>
7	Reset and reset cause	SW1 restarts the monitor; <code>SYS_RSTCAUSE</code> (0xF003) reports the expected source	monitor
8	LEDs	an application that toggles GPIOA[3:0] blinks D2–D5 in the same pattern (active low)	any <code>firmware/apps/</code> build
9	Application flash	monitor reads the JEDEC ID of U3; <code>make upload</code> writes and <code>v</code> verifies	monitor
10	A/B update	<code>make upload-slot SLOT=1 + make mon-boot + make commit</code> , then power-cycle and confirm the new image runs from slot B	host scripts
11	Rollback	upload a deliberately broken image to the free slot, boot it, reset — the loader must fall back to the other slot and set <code>BOOT_ERR[7]</code>	host scripts
12	Motor	scope on J6.3 shows the PWM waveform with the duty written to 0xF030; J6.4/J6.5 follow GPIOA[4]/[5]; pulling J6.6 low sets the fault bit	scope + <code>firmware/apps/motor_test.s</code>
13	Current measurement	record 1V1/2V5/3V3 current at 37.5 MHz with the monitor idle and with an application running (this closes OQ-B1)	bench supply

#	Step	Pass criterion	Tool
14	Long-run soak	24 h with the monitor running and the motor app cycling: no watchdog resets beyond those expected, no configuration re-entry	host log

11. Layout and EMC guidance

Topic	Guidance
Stack-up	4 layers, 1.6 mm: signal / solid GND / solid PWR (3V3, 1V1, 2V5 pours) / signal. A 2-layer board is possible but the 0.5 mm-pitch TQFP fan-out and the three rails make it unpleasant; not recommended
Fan-out	0.25 mm traces, 0.2 mm vias; TQFP-144 escape on all four sides. Ground pins 8/15/21/32/42/65/75/85/87/101/123/129/131/138 each get their own via to the GND plane
Decoupling	the 100 nF of C2–C20 within 3 mm of its pin, on the same side as the FPGA, with the via first at the pad; bulk (C21–C25) near the regulator outputs
Regulators	keep the two switchers' inductors (L1 33 μ H, L2 10 μ H), their catch diodes and the input loops tight — these are 150 kHz and ~500 kHz hard-switched nodes; keep L2 and D11 away from the configuration flash and the 25 MHz oscillator; put U5/U6/U7 and their electrolytics on the opposite edge of the board from J6
Motor return	J6 GND must have its own path to the input bulk capacitor; do not route motor current through the FPGA ground area. FB1 and the 3V3 bulk capacitor are the barrier
Oscillator	Y1 within 10 mm of pin 133, no via in the trace, ground pour under the part, keep the switching nodes > 5 mm away
Config flash	U2 within 15 mm of pins 46/47/49/54; the 100 Ω series resistors at the FPGA end (they are there to bound contention, §6.4)
USB	D+/D– as a 90 Ω differential pair, ESD device at the connector, CC1/CC2 5.1 k Ω within 10 mm of J8
JTAG	short TCK stub, no stubs beyond the two headers (TCK is the fastest signal at $\leq$ 25 MHz but the slowest is the cable); keep the TDO trace $\leq$ 50 mm
Test points	TP1–TP6 on the top side, labelled with the net names used in this document
Thermal	TQFP-144 has no exposed pad; use a copper pour on all four sides under the part, thermal vias under the body. Expected dissipation is well under 1 W (77 mA $\times$ 1.1 V core static plus switching), so no heatsink and no airflow requirement
Options/silkscreen	mark DNP parts (D6, J7, level shifter) and J5 pins 2/3/5/7 ("CFG FLASH — inputs only")

12. What this board turns the design into

MCU property	Where it comes from
Persistent program storage	U3 + sv16_flash_ctrl (MMIO 0xF0A0) + the bootloader reading A/B slots

MCU property	Where it comes from
Field reprogramming that cannot brick the board	A/B slots with PENDING/TRIED/GOOD records (ADR-019) + the monitor's upload/verify + JTAG as the last resort
Console / host interface	CH340C + UART0 at 115200 8N1, the same protocol <code>scripts/sv16_mon.py</code> speaks
Deterministic start-up	POR → configuration → startup FSM → 32768-cycle boot delay → boot attempt → monitor fallback (holding the RX line low for 4096 cycles forces the monitor, which is the escape hatch after a bad image)
Reset control with cause reporting	SW1 on <code>ext_rst_n</code> + <code>SYS_RSTCAUSE</code> at <code>0xF003</code>
Autonomous safety	watchdog at <code>0xF080</code> (<code>docs/PERIPHERALS.md</code>)
Digital I/O with a real expansion story	46 spare I/O on J7, 32 GPIO on J3/J4/J5 (2 ports × 16 bits at <code>0xF010/0xF070</code>), SPI0 at <code>0xF050</code>
Motor/actuator control	PWM at <code>0xF030</code> (DD-16 mA output on pin 88) with direction pins and a fault input
Debug	ROM monitor over UART today; the roadmap item S3 (UART GDB stub → JTAGG bridge, <code>REPORT.md</code> §8.2) uses this same JTAG header

13. Verification status and open items

13.1 Verified (with the artefact that proves it)

Item	Evidence
Pin numbers, pads, banks of all 52 constrained signals	<code>board/TQFP144_PINOUT.md</code> , generated from the prjtrellis device database for LFE5U-12F and the Lattice BSDL <code>PIN_MAP</code> ; <code>scripts/sv16_board_pins.py --check</code> fails on drift
Device pin budget (98 I/O, 11 dedicated, 6 VCC, 4 VCCAUX, 9 VCCIO, 14 GND, 2 NC = 144)	<code>board/ecp5_tqfp144.json</code> summary, validated by the script against both sources
Rail limits, ramp rates, POR trips, LVCMOS33 range	FPGA-DS-02012-3.4 Tables 3.1/3.2/3.3/3.4/3.5/3.11 (the PDF is in this repository)
Static supply currents	FPGA-DS-02012-3.4 Table 3.8
Configuration mode and pull-up values	TN1260 / TN-02039 (CFG[2:0] = 010) + TN-02038-2.0 Table 6.2 (4.7 kΩ PROGRAMN/INITN, 1 kΩ MCLK, 4.7–10 kΩ CSSPIN)
JTAG header pinout and TAP	ECP5 Versa EB98/EB103 Appendix A; BSDL <code>TAP_SCAN_*</code> attributes and <code>BOUNDARY_LENGTH</code> 409
Fit of the design in the device	<code>build/pll/sv16_nextpnr.log</code> : 9,191 LUT4, 4,772 FF, 18 DP16KD, 1 EHXPLL, 45.45 MHz

13.2 Open items (must be closed before or during layout)

ID	Item	Why it is open	Closing action
OQ-B1	Dynamic current on 1V1/2V5/3V3	the open-source flow has no power analyser; only datasheet static numbers exist	measure on the first board (\$10 step 13); the regulators are oversized on purpose

ID	Item	Why it is open	Closing action
OQ-B2	Regulator passives (L1 33 μ H, L2 10 μ H, the electrolytic values, the feedback divider)	taken from the vendors' typical application circuits rather than re-derived here; see also OQ-B10 for the values the two new bucks add	run the vendors' reference designs at layout time, and measure the rails at bring-up step 2
OQ-B3	Configuration time from U2 at the default MCLK	TN-02039 gives the mechanism but the bitstream's <code>MCLK_FREQ</code> setting is not inspected by the open flow	measure <code>INITN→DONE</code> with a scope on the first board
OQ-B4	CH340C 3.3 V connection details (V3 pin, decoupling, DTR polarity)	pin numbers and the 3.3 V configuration must come from the WCH datasheet, which is not in this repository	download and check the WCH datasheet before schematic capture
OQ-B5	USB-C sink compliance (Rd values, no-PD behaviour)	standard practice, no USB-IF testing	bench-test with several hosts/cables
OQ-B6	J2 (2×5 0.05") pin map versus the actual probe in use	conventions differ slightly between Digilent/FT2232H clones	check against the chosen adapter's cable
OQ-B7	EMC / ESD pre-compliance (motor transients, USB)	no test house, no hardware	add the layout measures of §11 and re-assess when hardware exists
OQ-B8	Mechanical outline, mounting, connector placement, enclosure	not fixed yet	decide with the mechanical drawing at PCB start
OQ-B9	J5 pin sharing with the configuration flash (§6.4)	acceptable as documented, but it is a usability wart	either keep the firmware rule or apply variant V1 (§14)
OQ-B10	Regulator passives taken from vendor typical applications: the MP1584 <code>COMP</code> network and <code>FREQ</code> resistor (R40), the AMS1117 output capacitor type (ceramic vs tantalum), and the exact <code>EN</code> threshold that sets the 13 ms delay (the delay scales as $-RC \cdot \ln(1 - V_{th}/V_{in})$)	the values here are correct in form but not re-derived from each datasheet	check the MP1584/AMS1117 application circuits before layout; measure the delayed rails at bring-up step 3
OQ-B11	Assembly of a 0.5 mm-pitch TQFP-144 plus 0603 passives without a reflow oven	this is the practical risk of the whole build, and it decides whether the first board works	order a stencil with the PCB, use solder paste + hot-air (₹2,500–4,000 stations are on Robu/Zbotic), drag-solder the TQFP with flux, or pay an Indian assembly service; practice on a ₹100 QFP breakout first
OQ-B12	USB-C (5 V) input voltage margin: 5 V – SS34 drop $\approx$ 4.6 V against a 4.5 V regulator minimum	the barrel jack is fine; USB is marginal under load	use 7–12 V on J9 for anything real, or replace D9 with a P-MOSFET ideal diode (drop $\approx$ 50 mV)

ID	Item	Why it is open	Closing action
OQ-B13	Actual FPGA stock at digikey.in / mouser.in / element14 India. Checked 27 Sep 2026: digikey.com listed 372 pieces live and a 40-week standard lead time — i.e. single pieces ship now, a back-order takes most of a year	the board is pointless without U1, and stock moves fast at this volume	check availability and price before ordering PCBs, and buy a spare; a speed-7 or industrial-grade part is an acceptable substitute (§3.4)
OQ-B14	Is Y1 an active oscillator? A 25 MHz HC49/US part is a 2-pin passive crystal and cannot drive <code>clk_25m</code> (the ECP5 has no on-chip crystal driver). Check the pin count before layout: 4 pins = active (works), 2 pins = passive (does not)	the board would configure but have no clock — no console, no boot	buy a 3.3 V active 25 MHz XO (3.2×2.5 or 5×7 mm), or build a Pierce oscillator around the crystal with an unbuffered inverter (e.g. 74LVC1G04 + 1 MΩ + 2 × 22 pF) — see PCB_COMPONENTS.md §10

14. Variants and the next revision

Variant	Change	Benefit	Cost
V1 — decouple J5 from the config flash	move <code>gpio_a[1], [2], [4], [6]</code> in <code>constraints/ecp5_144tqfp.lpf</code> from bank 8 (pins 46/47/49/51) to four of the free bank-2 pins (90-95), then regenerate <code>board/TQFP144_PINOUT.md</code>	J5 becomes 10 fully usable GPIO; the config flash no longer needs series resistors or a firmware rule; closes OQ-B9	one LPF edit, a bitstream rebuild, doc regeneration. Constraint changes touch the pin map docs and §6.4 only
V2 — single-cable USB	replace the CH340C with an FT232H (channel A = JTAG, channel B = UART)	one cable does configuration and console; no JTAG adapter needed	+USD 4-6, QFN-56 handling, two interfaces to document
V3 — on-board motor driver	populate U9 = DRV8871/TB6612 on the J6 nets, with current sense into a spare ADC-less comparator pin	turn-key actuator board	needs a driver design + thermal review
V4 — mixed I/O rails	split one bank to 1.8 V (e.g. for an SD card in bank 3)	expands the peripheral set	LPF <code>IOSTANDARD</code> change, an extra regulator, and the level-shift question returns
V5 — USB-C PD / PoE input	PD sink controller requesting 9-12 V, or 802.3af front end	no barrel jack	BOM + certification work

The order I would do them: **V1 first** (it is cheap and removes the only awkward constraint in this document), then V2 when the JTAG cable becomes the bottleneck, then V3 when a real motor is on the bench.

15. Appendix A — electrical quick reference (LFE5U-12F, from FPGA-DS-02012-3.4)

Parameter	Value	Table
VCC recommended / absolute max	1.045-1.155 V / -0.5...1.32 V	3.1, 3.2
VCCAUX recommended / absolute max	2.375-2.625 V / -0.5...2.75 V	3.1, 3.2
VCCIO recommended / absolute max	1.14-3.465 V / -0.5...3.63 V	3.1, 3.2

Parameter	Value	Table
LVC MOS33 VCCIO window	3.135–3.465 V	3.11
Supply ramp	0.01–10 V/ms, monotonic; VCCAUX $\leq$ 30 mV/ μ s	3.3
POR trip (VCC / VCCAUX / VCCIO8)	0.90–1.00 V / 2.00–2.20 V / 0.95–1.06 V; no VCCIO8 ramp-down detection	3.4
POR release condition	VCC and VCCAUX and VCCIO8 above trip	3.5
Static current, LFE5U-12F (TJ 85 °C, outputs tri-stated)	ICC 77 mA · ICCAUX 16 mA · ICCIO 0.5 mA/bank	3.8
Master SPI timing	fMCLK $\pm$ 20 %, duty 40–60 %; PROGRAMN low $\geq$ 110 ns ($\leq$ 50 ns is rejected); tINITL $\leq$ 55 μ s; tVMC $\leq$ 5 μ s; tCZ $\leq$ 300 ns	p85–87
JTAG TCK	$\leq$ 25 MHz (BSDL TAP_SCAN_CLOCK)	FPGA-MD-02097
Junction temperature	0–85 °C (commercial c grade)	3.2

16. Appendix B — files this document depends on

File	Role
board/ecp5_tqfp144.json	device pin database (built from the prjtrellis database + Lattice BSDL)
board/TQFP144_PINOUT.md	generated 144-pin net table — the netlist appendix of this document
scripts/sv16_board_pins.py	regenerates the table, checks it for drift, prints the bank summary
constraints/ecp5_144tqfp.lpf	the single source of truth for signal $\rightarrow$ pin (and for drive strength / pull mode)
docs/BOARD_AND_BRINGUP.md	bench notes, harness wiring and bring-up history for the simulation and previous hardware effort — read alongside §10
docs/SYNTHESIS_AND_DEPLOYMENT.md	build, timing, utilisation and existing pin-map narrative
docs/BOOT_AND_PROGRAMMING.md	image format, slot policy, monitor protocol, host tools
docs/MEMORY_MAP.md, docs/PERIPHERALS.md	MMIO map and register reference for every signal that reaches a connector
docs/MCU_READINESS.md	what is still missing for a production-grade MCU (the board does not change that list)
REPORT.md	status of the whole project, including the software-only roadmap §8.2
ECP5 and ECP5-5G.pdf	Lattice FPGA-DS-02012-3.4 — every rail, timing and pin-count number above

Revision 1.1 — **India-sourcing edition** (September 2026). The bill of materials was re-sourced so that every part is orderable from an Indian vendor: the two regulators became LM2596S-3.3 and MP1584EN, the LDO became AMS1117-2.5, the oscillator became the digikey.in YIC part, and the passives moved to 0603. Because the new bucks need $\geq$ 4.5 V input, the power-up order is now created by two RC-delayed enables instead of by deriving VCC/VCCAUX from 3V3 — §4.3 has the timeline, §3.6 the full change list, and §3.5 a module-first build for a first board. Prices checked against Indian vendor listings in September 2026.

Revision 1.0 — first issue, written against `constraints/ecp5_144tqfp.lpf` and the bitstream in `build/pll/sv16_top.bit`. No board exists yet; this document is the specification to build one.

LFE5U-12F-6TG144C — complete TQFP-144 pin / net table

Generated by scripts/
sv16_board_pins.py — **do not edit by hand**. Device data: board/ecp5_tqfp144.json; design signals: constraints/
ecp5_144tqfp.lpf; board destinations: BOARD.md §5–§7. Part: **LFE5U-12F-6TG144C** (TQFP144 (20 x 20 mm, 0.5 mm
pitch)).

1. I/O bank summary

Bank	VCCIO rail	I/O pins	Used by this design	Free
0	3V3 (pin 137)	10	10	0
1	3V3 (pin 122)	16	12	4
2	3V3 (pin 100)	18	12	6
3	3V3 (pin 70, 86)	15	2	13
6	3V3 (pin 16, 36)	15	0	15
7	3V3 (pin 9)	12	4	8
8	3V3 (pin 43)	12	12	0

Total: 98 user I/O, 52 used by the design, 46 free.

2. Configuration-pad interactions (CFG[2:0] = 010, Master SPI)

Pin	Pad	sysCONFIG role	LPF signal	Consequence on the board
39	PB6A	D5/IO5	led[0]	parallel/slave-mode pad only, free in this configuration
40	PB4A	D7/IO7	led[1]	parallel/slave-mode pad only, free in this configuration
41	PB6B	D4/IO4	led[2]	parallel/slave-mode pad only, free in this configuration
44	PB4B	D6/IO6	led[3]	parallel/slave-mode pad only, free in this configuration
45	PB9A	D3/IO3	gpio_a[0]	parallel/slave-mode pad only, free in this configuration
46	PB11A	D1/MISO (Master SPI pad)	gpio_a[1]	shares the net with the config flash — see BOARD.md §6.4
47	PB11B	D0/MOSI (Master SPI pad)	gpio_a[2]	shares the net with the config flash — see BOARD.md §6.4
48	PB13A	CSN/SN	gpio_a[3]	parallel/slave-mode pad only, free in this configuration
49	PB15A	HOLDN/DI/BUSY/CSSPIN (Master SPI pad)	gpio_a[4]	shares the net with the config flash — see BOARD.md §6.4

Pin	Pad	sysCONFIG role	LPF signal	Consequence on the board
50	PB13B	CS1N	gpio_a[5]	parallel/slave-mode pad only, free in this configuration
51	PB15B	DOUT/CSON (Master SPI pad)	gpio_a[6]	shares the net with the config flash — see BOARD.md §6.4
52	PB18A	WRITEN	gpio_a[7]	parallel/slave-mode pad only, free in this configuration

Pins sharing a net with the configuration flash: 46, 47, 49, 51.

3. Complete pin table (1-144)

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
1	PL5C	io	bank 7 (3V3)	gpio_b[12]	8 mA	J3.15 EXP-B header
2	PL5D	io	bank 7 (3V3)	gpio_b[13]	8 mA	J3.16 EXP-B header
3	PL5B	io	bank 7 (3V3)	gpio_b[14]	8 mA	J3.17 EXP-B header
4	PL8A	io	bank 7 (3V3)	gpio_b[15]	8 mA	J3.18 EXP-B header
5	PL8C	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
6	PL8B	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
7	PL8D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
8	VSSIO7	gnd	GND plane	—	—	GND plane
9	VCCIO7	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
10	PL14A	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
11	PL14C	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
12	PL14B	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
13	PL14D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout
14	PL23D	io	bank 7 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
15	VSSIO6	gnd	GND plane	—	—	GND plane
16	VCCIO6	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
17	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
18	PL32B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
19	PL35B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
20	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
21	VSS	gnd	GND plane	—	—	GND plane
22	PL35C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
23	PL35D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
24	PL38A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
25	PL38B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
26	PL41C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
27	PL41D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
28	PL44A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
29	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
30	PL44B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
31	PL44C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
32	VSSIO6	gnd	GND plane	—	—	GND plane
33	PL44D	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
34	PL47A	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
35	PL47B	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
36	VCCIO6	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
37	PL47C	io	bank 6 (3V3)	—	—	spare I/O — route to J7 breakout
38	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
39	PB6A	io	bank 8 (3V3)	led[0]	8 mA	D2 LED (active low) + R31 470R
40	PB4A	io	bank 8 (3V3)	led[1]	8 mA	D3 LED (active low) + R32 470R
41	PB6B	io	bank 8 (3V3)	led[2]	8 mA	D4 LED (active low) + R33 470R
42	VSSIO8	gnd	GND plane	—	—	GND plane
43	VCCIO8	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
44	PB4B	io	bank 8 (3V3)	led[3]	8 mA	D5 LED (active low) + R34 470R
45	PB9A	io	bank 8 (3V3)	gpio_a[0]	8 mA	J4.1 EXP-C header
46	PB11A	io	bank 8 (3V3)	gpio_a[1]	8 mA	J4.2 EXP-C header
47	PB11B	io	bank 8 (3V3)	gpio_a[2]	8 mA	J4.3 EXP-C header
48	PB13A	io	bank 8 (3V3)	gpio_a[3]	8 mA	J4.4 EXP-C header
49	PB15A	io	bank 8 (3V3)	gpio_a[4]	8 mA	J4.5 EXP-C header
50	PB13B	io	bank 8 (3V3)	gpio_a[5]	8 mA	J4.6 EXP-C header
51	PB15B	io	bank 8 (3V3)	gpio_a[6]	8 mA	J4.7 EXP-C header
52	PB18A	io	bank 8 (3V3)	gpio_a[7]	8 mA	J4.8 EXP-C header
53	VCCAUX	vcclaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
54	CCLK	config	config	—	—	U5.6 config flash CLK (R7 100R series, R8 1k pull-up to VCCIO8)
55	INITN	config	config	—	—	J1.10 + R9 4.7k pull-up to VCCIO8 + D6 red LED via R15 1k
56	DONE	config	config	—	—	J1.9 + R10 4.7k pull-up to VCCIO8 + D1 green LED via R32 1k
57	PROGRAMN	config	config	—	—	J1.4 + R11 4.7k pull-up to VCCIO8 + SW2 + Q1 (DTR auto-program, JP1 to disable)
58	CFG_2	config	config	—	—	R14 1k to GND (CFGMDN2 = 0)
59	CFG_1	config	config	—	—	R13 4.7k to VCCIO8 (CFGMDN1 = 1)
60	TDO	jtag	JTAG	—	—	J1.2 (TDO)
61	TDI	jtag	JTAG	—	—	J1.3 (TDI)
62	CFG_0	config	config	—	—	R12 1k to GND (CFGMDN0 = 0)
63	TCK	jtag	JTAG	—	—	J1.8 (TCK)
64	TMS	jtag	JTAG	—	—	J1.6 (TMS)
65	VSS	gnd	GND plane	—	—	GND plane
66	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
67	PR47B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
68	PR47D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
69	PR47C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
70	VCCIO3	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
71	PR47A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
72	PR44C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
73	PR41D	io	bank 3 (3V3)	uart_rx	4 mA / UP	U1 TXD via R21 0R / J5.3 via R22 0R
74	PR41C	io	bank 3 (3V3)	uart_tx	4 mA	U1 RXD via R23 0R / J5.2 via R24 0R
75	VSSIO3	gnd	GND plane	—	—	GND plane
76	PR41B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
77	PR41A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
78	PR35D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
79	PR35B	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
80	PR35C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
81	PR35A	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
82	PR32D	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
83	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
84	PR32C	io	bank 3 (3V3)	—	—	spare I/O — route to J7 breakout
85	VSS	gnd	GND plane	—	—	GND plane
86	VCCIO3	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
87	VSSIO3	gnd	GND plane	—	—	GND plane
88	PR23D	io	bank 2 (3V3)	pwm_out	16 mA	J6.3 motor connector
89	PR23B	io	bank 2 (3V3)	motor_dir1	8 mA	J6.4 motor connector
90	PR23C	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
91	PR23A	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
92	PR20D	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
93	PR20B	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
94	PR20C	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
95	PR20A	io	bank 2 (3V3)	—	—	spare I/O — route to J7 breakout
96	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
97	PR14B	io	bank 2 (3V3)	gpio_a[8]	8 mA	J2.3 EXP-A header
98	PR14C	io	bank 2 (3V3)	gpio_a[9]	8 mA	J2.4 EXP-A header
99	PR14A	io	bank 2 (3V3)	gpio_a[10]	8 mA	J2.5 EXP-A header
100	VCCIO2	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
101	VSSIO2	gnd	GND plane	—	—	GND plane
102	PR8D	io	bank 2 (3V3)	motor_dir2	8 mA	J6.5 motor connector
103	PR8B	io	bank 2 (3V3)	motor_fault_n	4 mA / UP	J6.6 motor connector
104	PR8C	io	bank 2 (3V3)	gpio_a[11]	8 mA	J2.6 EXP-A header
105	PR8A	io	bank 2 (3V3)	gpio_a[12]	8 mA	J2.7 EXP-A header
106	PR2D	io	bank 2 (3V3)	gpio_a[13]	8 mA	J2.8 EXP-A header
107	PR2C	io	bank 2 (3V3)	gpio_a[14]	8 mA	J2.9 EXP-A header
108	PR2B	io	bank 2 (3V3)	gpio_a[15]	8 mA	J2.10 EXP-A header
109	NC	nc	do not connect	—	—	do not connect
110	PT67A	io	bank 1 (3V3)	flash_sck	8 mA	U6.6 application flash SCK
111	PT62B	io	bank 1 (3V3)	flash_cs_n	8 mA	U6.1 application flash /CS

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
112	PT62A	io	bank 1 (3V3)	flash_mosi	8 mA	U6.5 application flash DI
113	PT60B	io	bank 1 (3V3)	flash_miso	4 mA / UP	U6.2 application flash DO
114	PT60A	io	bank 1 (3V3)	spi0_sck	8 mA	J2.12 SPI0 expansion
115	PT58B	io	bank 1 (3V3)	spi0_cs_n	8 mA	J2.11 SPI0 expansion
116	PT58A	io	bank 1 (3V3)	spi0_mosi	8 mA	J2.13 SPI0 expansion
117	PT56B	io	bank 1 (3V3)	spi0_miso	4 mA / UP	J2.14 SPI0 expansion
118	PT56A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
119	PT53B	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
120	PT53A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
121	PT51A	io	bank 1 (3V3)	—	—	spare I/O — route to J7 breakout
122	VCCIO1	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
123	VSSIO1	gnd	GND plane	—	—	GND plane
124	PT42B	io	bank 1 (3V3)	gpio_b[8]	8 mA	J3.11 EXP-B header
125	PT42A	io	bank 1 (3V3)	gpio_b[9]	8 mA	J3.12 EXP-B header
126	PT40B	io	bank 1 (3V3)	gpio_b[10]	8 mA	J3.13 EXP-B header
127	PT40A	io	bank 1 (3V3)	gpio_b[11]	8 mA	J3.14 EXP-B header
128	PT29A	io	bank 0 (3V3)	gpio_b[7]	8 mA	J3.10 EXP-B header
129	VSSIO0	gnd	GND plane	—	—	GND plane
130	VCC	vcc	1V1 core rail (U3 switcher)	—	—	1V1 core rail (U3 switcher) — 4x 100n + 10u
131	VSS	gnd	GND plane	—	—	GND plane
132	VCCAUX	vccaux	2V5 rail (U4 LDO)	—	—	2V5 rail (U4 LDO) — 100n + 4u7
133	PT15B	io	bank 0 (3V3)	clk_25m	4 mA	Y1.3 25 MHz XO output

Pin	Pad	Kind	Bank / rail	Design signal	Drive / pull	Board net
134	PT15A	io	bank 0 (3V3)	ext_rst_n	4 mA / UP	SW1 reset button (to GND) + R1 10k pull-up + C1 100n
135	PT13B	io	bank 0 (3V3)	gpio_b[0]	8 mA	J3.3 EXP-B header
136	PT13A	io	bank 0 (3V3)	gpio_b[1]	8 mA	J3.4 EXP-B header
137	VCCIO0	vccio	3V3 rail	—	—	3V3 rail — 100n per pin
138	VSSIO0	gnd	GND plane	—	—	GND plane
139	PT9B	io	bank 0 (3V3)	gpio_b[2]	8 mA	J3.5 EXP-B header
140	PT9A	io	bank 0 (3V3)	gpio_b[3]	8 mA	J3.6 EXP-B header
141	PT6B	io	bank 0 (3V3)	gpio_b[4]	8 mA	J3.7 EXP-B header
142	PT4B	io	bank 0 (3V3)	gpio_b[5]	8 mA	J3.8 EXP-B header
143	PT6A	io	bank 0 (3V3)	gpio_b[6]	8 mA	J3.9 EXP-B header
144	NC	nc	do not connect	—	—	do not connect

Pad names and the dedicated sysCONFIG/TAP pins are from Lattice BSDL FPGA-MD-02097 (LFE5U_25F_XXTG144); the LFE5U-12F/25F/45F 144-TQFP pinout is identical (FPGA-DS-02012-3.4 Table 1.1 lists 98 user I/O on 144 TQFP for each density). The 98 user-I/O pin numbers and their banks are taken from the prjtrellis database for LFE5U-12F and agree with the BSDL.